



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PURE MATHEMATICS
PAPER 2

6042/2
3 hours

JUNE 2025 SESSION

Additional materials:
Answer booklet
List of Formulae MF 7
Scientific calculator [Non-programmable]
Graph paper

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer all questions in Section A and any five questions from Section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 5 printed pages.

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CZJMS/EC 12025

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Section A (40 marks).

Answer all questions in this section.

- 1 (a) Sketch the graph of $y = x^2 - 2x - 3$, clearly showing the intercepts and coordinates. [3]
- (b) Hence or otherwise, find the area enclosed by the curve, the axes and the points $x = 3$ and $x = -1$. [3]
- 2 Relative to the origin O the position vectors of the points P , Q and R are given by $\overrightarrow{OP} = \begin{pmatrix} 4 \\ 10 \\ -6 \end{pmatrix}$; $\overrightarrow{OQ} = \begin{pmatrix} 2 \\ 8 \\ -4 \end{pmatrix}$; $\overrightarrow{OR} = \begin{pmatrix} 6 \\ 6 \\ -4 \end{pmatrix}$
- (a) Find $\overrightarrow{PQ}$. [2]
- (b) Find the angle PQR . [4]
- 3 The complex numbers Z and W are such that $W = a + 2i$ and $Z = -1 - 3bi$ where a and b are real and positive constants.
- Given that $WZ = 5 - 8i$, find the exact values of a and b . [7]
- 4 The first, second and third terms of a geometric progression are the first, ninth and twenty-first terms of an arithmetic progression respectively.
- The first term of each progression is 4 and the common ratio of the geometric progression is r , where $r \neq 1$.
- Find the
- (a) value of r , [4]
- (b) 4th term of each progression. [4]
- 5 If $x = 4t(1 - t^2)$ and $y = 1 - t^2$.
- (a) Find $\frac{dy}{dx}$. [3]
- (b) Find the coordinates of the point where
- (i) the gradient is zero, [2]
- (ii) the gradient is undefined. [2]
- (c) Find the equations of the tangents at the origin. [3]
- (d) Find the equations of the normal at the origin. [3]

Section B (80 marks)

Answer any five questions from this section. Each question carries 16 marks.

6 (a) Find the values of x for which $\begin{vmatrix} x-1 & 2x & x+1 \\ x & x+1 & x-1 \\ 1 & 1 & 1 \end{vmatrix} = 0$. [2]

(b) Matrices P and Q are such that $P = \begin{pmatrix} 0 & 2 & 3 \\ 2 & 0 & 0 \\ 1 & -1 & 0 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$

Find

(i) PQ [2]

(ii) $(PQ)^{-1}$ [7]

(c) Hence find the values of x , y and z for which $PQ \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 23 \\ 6 \\ 5 \end{pmatrix}$ [5]

7 (a) Express $\frac{2}{(1+x)(1+3x)}$ in partial fractions. [5]

(b) Hence or otherwise solve the differential equation

$$\frac{dy}{dx} = \frac{2(y+2)}{(1+x)(1+3x)}$$

given that $y = -\frac{1}{3}$ when $x = 2$, expressing y in terms of x . [11]

8 The function $f: x \mapsto -3x^2 + 2x + 1$, $x \geq P$.

(a) Express $f(x)$ in the form $A(x+B)^2 + C$, where A , B and C are constants. [3]

(b) State the value of P for which $f(x)$ has an inverse. [1]

(c) With the aid of diagrams, describe the geometrical transformations that map the graph of $y = x^2$ onto the graph of $y = f(x)$. [12]

9 (a) (i) Prove the identity

$$\sin(\theta - 30^\circ) + \cos(\theta - 60^\circ) = \sqrt{3}\sin\theta$$
 [3]

(ii) Hence solve the equation

$$\sin(\theta - 30^\circ) + \cos(\theta - 60^\circ) = \frac{1}{2}\sec\theta,$$

in the range $0 \leq \theta \leq 360^\circ$. [6]



- (b) It is given that $a = \sin \theta - 2 \cos \theta$ and $b = 2 \sin \theta + \cos \theta$,
where $0 \leq \theta \leq 360^\circ$.

(i) Show that $a^2 + b^2$ has a constant value for all values of θ . [3]

(ii) Find the values of θ for which $4a = b$. [4]

- 10 (a) Set $S = \{p, q, r, v\}$. The operation $\circ$ is performed on S with the following results: $p \circ x = x$, $q \circ q = p$, $q \circ r = v$, $q \circ v = r$, $r \circ r = p$, $v \circ v = p$.

(i) Construct a Cayley table using the given information. [4]

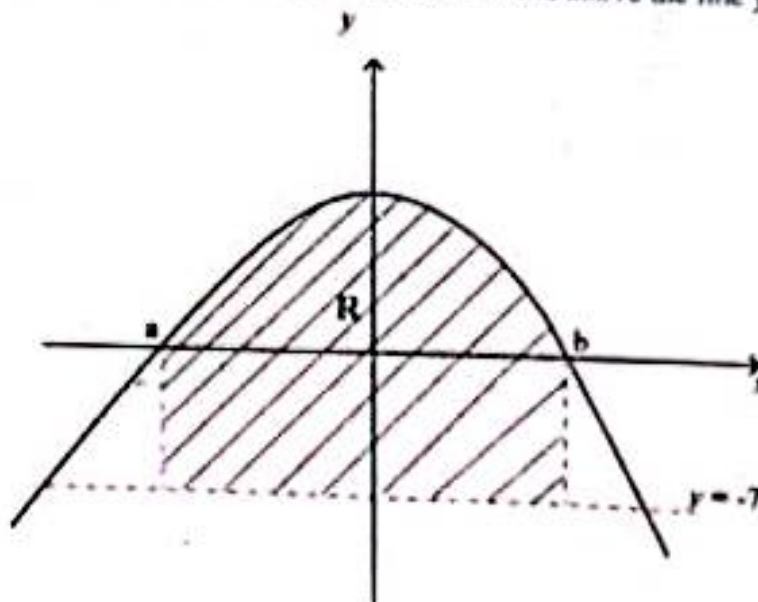
(ii) Show that S is a group under $\circ$. [6]

- (b) Prove by induction that, for all integers $n \geq 1$, $3^n + 1$, is divisible by 2. [6]

- 11 For this question, x is small for x^3 and higher powers, to be ignored.

(a) Show that $\cos 2x = 1 - 2x^2$. [2]

- (b) The diagram shows the graph of $f(x) = \cos 2x$ and graph of $y = -7$. The shaded region, R , (shown on the graph) is the region below the graph $y = 1 - 2x^2$, between $x = a$ and $x = b$ and above the line $y = -7$.



(i) Find the roots of the equation $1 - 2x^2 = 0$. [2]

(ii) Calculate the area of region R . [3]

(iii) Use the Trapezium Rule with 4 stripes to estimate the area of region, R . [7]



(iv) Calculate the percentage error in calculating area of region R, using the Trapezium Rule. [2]

12 (a) Find the sum of all integers between 99 and 405 that are divisible by 7. [4]

(c) (i) Express $\frac{5r+4}{r(r+1)(r+2)}$ in partial fractions. [4]

(i) Hence prove by the method of differences that

$$\sum_{r=1}^n \frac{5r+4}{r(r+1)(r+2)} = \frac{n(7n+11)}{2(n+1)(n+1)} \quad [5]$$

(ii) Find the value of

$$\sum_{r=1}^{10} \frac{5r+4}{r(r+1)(r+2)}$$

as a recurring decimal. [3]

